

Symbolic Compression Protocols for Multilingual Multi-Agent Handoff Across Frontier Models

Anonymous Authors

Abstract

Large language model agents increasingly need to transfer operational state across models, providers, and languages without carrying full conversational histories. We study compact symbolic formats as *state-transfer protocols*, asking when they preserve semantic fidelity, operational continuity, and handoff quality across heterogeneous model families. EXP05 is a broad discovery experiment: it compares natural, compressed, hybrid_min, and hybrid_state protocols over Spanish, English, and Chinese tasks using multiple frontier and semi-open generators, plus judge-drift, token-budget, Chinese compactness, and long-handoff mini-experiments. EXP06 is a controlled follow-up: it directly pairs compressed against hybrid_state, adds a Chinese tokenizer/variant ablation, and evaluates every valid generation with three judges. EXP07 and EXP08 add real-agent validation layers using LangGraph and OpenFang with objective handoff metrics; EXP08 scales this setting to 900 operational executions. Across EXP05, compressed is the strongest general baseline, while hybrid_state significantly improves paired state preservation in the principal subset ($\Delta = +0.133$, 95% CI $[+0.025, +0.240]$). EXP06 confirms the state-preservation advantage under stricter controls ($\Delta = +0.066$, 95% CI $[+0.032, +0.100]$), but also shows that hybrid_state does not win global quality ($\Delta Q = -0.208$, 95% CI $[-0.239, -0.177]$). EXP07 and EXP08 confirm that these protocols can be evaluated in real agent surfaces without relying solely on LLM judges, but neither shows a global hybrid_state win: compressed remains the strongest operational baseline, while hybrid_state shows only conditional advantages in selected state/constraint strata. EXP08 also identifies a provider/model failure mode: under strict JSON requirements, Azure GPT 5.4 High with high reasoning produced 76 blank-output cells, especially for hybrid_state. We therefore argue that compressed is the universal baseline, while hybrid_state is a specialized protocol whose benefits are strongest for explicit state variables, not aggregate quality. Chinese and tokenizer-ablation results suggest language representation matters, but the mechanism remains only partially isolated.

1 Introduction

Autonomous and semi-autonomous language-model agents often need to continue work after context truncation, model substitution, provider failure, or handoff to another agent. The usual solution is to retain more natural-language context. This is expensive and fragile: longer context can increase cost, latency, and ambiguity, while still failing to preserve the operational state that matters for continuation.

This paper studies a different framing: compressed and symbolic prompt formats as *state-transfer protocols*. Rather than asking only whether a prompt can be shortened, we ask whether compact formats preserve the information required for another model to continue a task. This distinction matters because a protocol can be token-efficient yet operationally weak, or slightly less natural yet better at carrying state.

The empirical design has four stages. EXP05, named EXP05_PREMIUM_MULTIMODEL_TRILINGUAL, is a broad discovery experiment across modes, languages, model families, judges, and stress tests.

EXP06, named `EXP06_CAUSAL_PROTOCOL_TRANSFER`, is a narrower confirmation experiment that directly tests the strongest EXP05 claim: `hybrid_state` should improve operational state transfer even if it does not maximize global quality. EXP07, named `EXP07_REAL_AGENT_HANDOFF_OBJECTIVE_METRICS`, asks whether the same protocol family can be evaluated inside real agent frameworks using objective continuation metrics rather than only LLM judges. EXP08, named `EXP08_REAL_AGENT_SCALEUP`, scales the real-agent evaluation to 30 tasks, 5 repetitions, and 900 operational executions.

The resulting story is deliberately not a winner-takes-all ranking. Compressed text remains the strongest universal baseline. `Hybrid_state` is weaker on aggregate quality, but it repeatedly improves state preservation in judge-based experiments, the metric most aligned with multi-agent continuation. EXP07 and EXP08 further show that real-agent objective metrics are feasible and useful, while also warning against overgeneralization: `hybrid_state` does not dominate real-agent execution globally. This distinction is central: compact protocols should be optimized against their intended transfer function, not only against a single global score.

2 Related Work

This work connects four lines of research.

First, prompt compression studies attempt to reduce token count while preserving task utility. These approaches usually optimize for shorter prompts, shorter context, or cheaper inference. EXP05 instead treats compact prompts as communication protocols between agents.

Second, work on LLM agents and tool-using systems emphasizes memory, planning, and handoff. Many systems use natural-language summaries as persistent state. Our experiment asks whether more structured symbolic forms can preserve operational state more reliably.

Third, multilingual evaluation shows that model behavior varies by language, tokenization, training distribution, and cultural-linguistic structure. EXP05 directly compares Spanish, English, and Chinese to test whether compact symbolic formats behave differently across scripts and tokenization regimes.

Fourth, LLM-as-judge methodology has become common for scalable evaluation, but judge drift remains a serious concern. We therefore include a dedicated judge-drift mini-experiment comparing Gemini, GPT, and Grok judges over the same outputs.

3 Method

3.1 Communication Modes

We compare four protocol modes.

natural uses standard human-readable prose.

compressed uses a compact baseline similar to dense operational notes.

hybrid_min uses the most aggressively minimized hybrid symbolic form.

hybrid_state emphasizes explicit state preservation, continuity markers, and handoff-relevant fields.

3.2 Languages and Tasks

The principal experiment uses Spanish (ES), English (EN), and Chinese (ZH). Tasks cover contextual summarization, handoff, planning, persistent operational state, and multilingual transfer. The principal design contains 15 task templates, 3 languages, 4 modes, 2 repetitions, and 5 generator families, partitioned into resumable execution parts.

3.3 Models

The principal generator pool includes Grok 4.20 Non-Reasoning, Gemini 3.1 Pro Preview, Llama 4 Maverick, Qwen3-Next-80B, and GLM 5. Additional controlled extensions include Azure GPT instant, Grok reasoning/non-reasoning comparisons, Azure Grok 4.3, judge-drift models, token-budget stress models, and long-handoff models.

3.4 Evaluation

Evaluators return strict JSON metrics. Positive metrics are scored with higher values better: semantic fidelity, clarity, utility, completeness, state preservation, operational continuity, context recoverability, handoff quality, compactness, language stability, and inter-model transferability. Negative metrics are scored with lower values better: ambiguity and information loss.

We compute a descriptive quality index:

$$Q = \frac{\text{mean(positive)} + (6 - \text{mean(negative)})}{2}.$$

This index is used for ranking, but individual metrics are reported separately because the main scientific claims concern state preservation, continuity, compactness, and language behavior.

3.5 Data Cleaning

All analyses use cleaned and deduplicated outputs. The cleaning pass keeps the latest successful generation/evaluation per cell, removes superseded retries, preserves unique MiniEXP-C generation failures as experimental evidence, and verifies that no evaluation points to a missing generation. EXP06 additionally preserves provider policy failures as terminal events: 21 Azure GPT instant generation cells were marked `policy_failure`, retained for provider-reliability analysis, and excluded from score aggregation because no output existed to evaluate. EXP07 preserves operational startup/calibration errors separately, then scores the latest successful execution per real-agent cell. EXP08 reports two views: an *operational-complete* view with 900/900 latest cells marked OK, and a stricter *JSON-valid* view with 824/900 cells after excluding blank-output responses from Azure GPT 5.4 High.

4 Experiments

4.1 Principal EXP05

The principal experiment evaluates the four communication modes across ES/EN/ZH and five generator families. The cleaned principal subset contains 1,800 evaluations.

4.2 MiniEXP-A: Grok Reasoning

MiniEXP-A compares Grok 4.20 Non-Reasoning and Grok 4.20 Reasoning on matched compact-protocol tasks. A2 separately evaluates Azure Grok 4.3 and is not pooled with Grok 4.20.

4.3 MiniEXP-B: Judge Drift

MiniEXP-B re-evaluates the same 120 source outputs with four judges: Gemini 3.1 Pro Preview, Gemini 3.5 Flash, Azure GPT 5.4 High, and Grok 4.20 Reasoning.

4.4 MiniEXP-C: Token Budget Stress Test

MiniEXP-C evaluates how models respond to reduced output budgets. Generation failures are not discarded when no successful retry exists, because they indicate stress-test failure.

4.5 MiniEXP-D: Chinese Compactness

MiniEXP-D isolates Chinese compact protocol behavior to test whether ZH performance reflects a real language/tokenization signal or only principal-experiment noise.

4.6 MiniEXP-E: Long Handoff

MiniEXP-E studies multi-turn handoff durability under longer operational chains.

4.7 EXP06: Controlled Causal Follow-up

EXP06 tests the strongest EXP05 interpretation under stricter controls. EXP06-A directly pairs compressed and hybrid_state on causal handoff tasks across ES, EN, and ZH. EXP06-B probes the language/tokenization hypothesis using ZH, ZH_PINYIN, EN_LITERAL, and ES_LITERAL variants. EXP06-C evaluates every valid generation with three independent judges: Gemini 3.5 Flash, Azure GPT 5.4 High, and Grok 4.20 Reasoning. The cleaned EXP06 dataset contains 1,563 valid generations and 4,689 successful evaluations.

4.8 EXP07: Real-Agent Handoff

EXP07 evaluates the same protocol family inside real agent surfaces. It uses LangGraph StateGraph and OpenFang CLI/daemon runs over 10 operational handoff tasks, 3 modes, and 3 repetitions. LangGraph is run with Azure GPT chat latest and Azure GPT 5.4 High; OpenFang is run with Azure GPT chat latest. The cleaned EXP07 dataset contains 270 successful real-agent executions after deduplication, with 4 operational calibration errors retained separately. Unlike EXP05 and EXP06, EXP07 uses objective schema metrics: variable recovery, subtask completion, plan continuity, constraint retention, handoff success, and state-error count.

4.9 EXP08: Real-Agent Scale-Up

EXP08 scales the real-agent design to 30 operational tasks, 2 modes, 5 repetitions, and 900 operational executions. LangGraph is run with Azure GPT chat latest and Azure GPT 5.4 High; OpenFang is run with Azure GPT chat latest. The goal is not to introduce a new protocol, but to test whether EXP07’s real-agent conclusions survive a larger matrix. The final operational view contains 900/900 OK cells and zero latest operational errors. The stricter JSON-valid view contains 824/900 cells; the remaining 76 are blank-output responses from Azure GPT 5.4 High under LangGraph, concentrated more heavily in hybrid_state.

5 Results

5.1 Principal Mode Results

Table 1 shows the principal experiment ranking by mode. Compressed achieves the highest quality index. Hybrid_state ranks second and is strongest on state preservation.

Mode	n	Q	Fidelity	State	Continuity	Compact	InfoLoss
compressed	450	4.301	4.351	4.233	4.327	4.833	1.949
hybrid_state	450	4.180	4.189	4.369	4.320	4.807	2.022
natural	450	4.161	4.184	3.991	4.024	4.409	2.033
hybrid_min	450	3.769	3.853	3.840	3.742	4.849	2.467

Table 1: Principal EXP05 results by communication mode. Negative metrics such as information loss are better when lower.

EXP05 principal modes

Compressed leads global quality; hybrid_state leads state preservation; hybrid_min mainly preserves compactness.

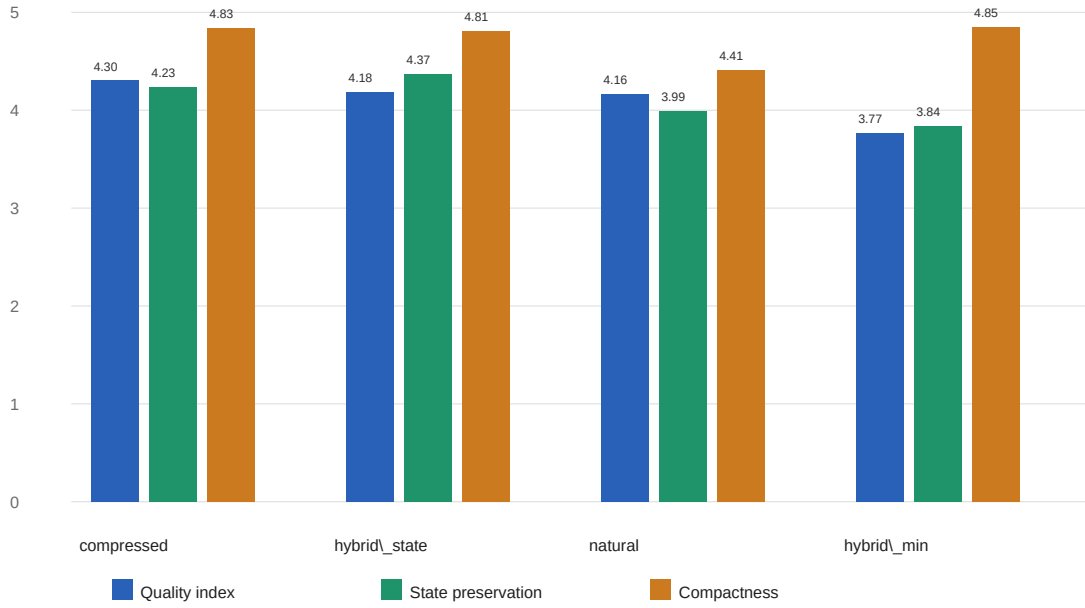


Figure 1: EXP05 principal mode tradeoff across quality index, state preservation, and compactness.

5.2 Paired Effects Against Compressed

Table 2 reports paired deltas against compressed. Hybrid_state improves state preservation by +0.133 with 95% CI [+0.025, +0.240]. Its operational continuity is effectively tied with compressed. Hybrid_min preserves compactness but loses quality, fidelity, state, and continuity.

5.3 Language Effects

Table 3 shows a consistent ZH advantage in the principal subset. This supports the hypothesis that language structure and tokenization may affect compact protocol performance, but it does not yet isolate the causal mechanism.

Metric	Mode	n	Δ	95% CI Low	95% CI High
quality index	hybrid_state	430	-0.108	-0.203	-0.012
semantic fidelity	hybrid_state	430	-0.142	-0.255	-0.029
state preservation	hybrid_state	430	+0.133	+0.025	+0.240
operational continuity	hybrid_state	430	+0.007	-0.099	+0.113
compactness	hybrid_state	430	-0.014	-0.080	+0.052

Table 2: Paired deltas for hybrid_state against compressed in the principal experiment.

Language	n	Q	Fidelity	State	Continuity
ZH	600	4.180	4.207	4.197	4.180
EN	600	4.108	4.143	4.115	4.108
ES	600	4.020	4.083	4.013	4.022

Table 3: Principal EXP05 results by language.

5.4 Generator Effects

Grok 4.20 Non-Reasoning leads the principal generator ranking ($Q = 4.466$), followed by GLM 5 ($Q = 4.394$), Qwen3-Next-80B ($Q = 4.248$), Llama 4 Maverick ($Q = 3.874$), and Gemini 3.1 Pro Preview ($Q = 3.532$). This ranking should be interpreted as protocol/task behavior under this setup, not as a general benchmark of model intelligence.

5.5 Judge Drift

Judge-drift results show strong but imperfect agreement. Gemini 3.1 and Gemini 3.5 correlate most strongly ($r = 0.839$). Azure GPT 5.4 High is consistently stricter than Gemini and Grok judges, with mean quality-index deltas around -0.95 versus Gemini and -0.43 versus Grok.

5.6 Token Budget Stress

Gemini, Grok, and Qwen remain structurally successful at all tested budgets, including 35%. Azure GPT instant is fragile under aggressive budgets: success is 47.9% at 35%, 77.1% at 50%, 95.8% at 75%, and 97.9% at 100%.

6 Controlled Follow-up: EXP06

EXP05 establishes a broad multilingual and multi-model signal: compressed protocols are a strong baseline, while hybrid_state improves state preservation. EXP06 tests this claim under stricter paired controls. The experiment removes natural and hybrid_min, directly compares compressed against hybrid_state, and adds a Chinese tokenizer/variant ablation plus three-judge calibration.

6.1 Design

EXP06 contains two generation phases and one evaluation phase. EXP06-A evaluates causal hand-off transfer across ES, EN, and ZH. EXP06-B evaluates ZH, ZH_PINYIN, EN_LITERAL, and ES_LITERAL variants to probe the language/tokenization hypothesis. EXP06-C evaluates every valid generation with Gemini 3.5 Flash, Azure GPT 5.4 High, and Grok 4.20 Reasoning. The cleaned

dataset contains 1,563 valid generations and 4,689 evaluations; 21 Azure GPT instant generations were retained as terminal policy failures and excluded from scoring.

6.2 Mode Results

Compressed achieves the higher global quality index in EXP06 (Table 4). Hybrid_state is weaker on global quality, semantic fidelity, operational continuity, compactness, ambiguity, and information loss. However, it is stronger on state preservation. This is the central controlled result: EXP06 confirms hybrid_state as a state-transfer specialization, not as a universal quality winner.

Mode	n	Q	Fidelity	State	Continuity	Compact	Ambig.	InfoLoss
compressed	777	3.855	3.780	3.728	3.761	4.705	2.231	2.498
hybrid_state	786	3.661	3.621	3.805	3.699	4.638	2.678	2.642

Table 4: EXP06 mode-level results. Higher is better for Q , fidelity, state, continuity, and compactness. Lower is better for ambiguity and information loss.

EXP06 controlled comparison

EXP06 confirms the tradeoff: compressed wins global quality; hybrid_state preserves slightly more state.

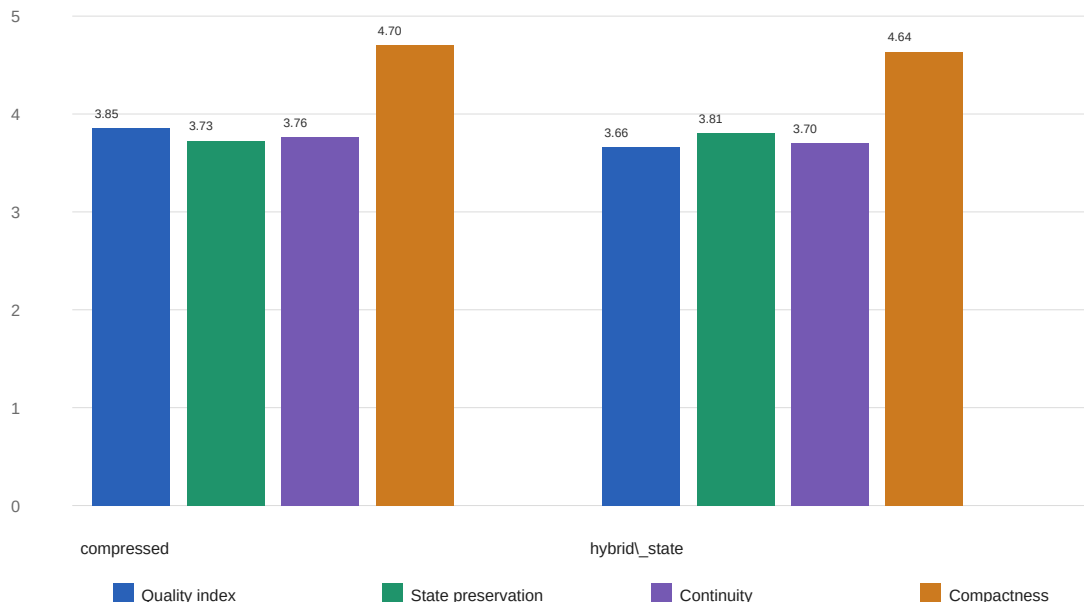


Figure 2: EXP06 controlled comparison between compressed and hybrid_state. The figure highlights the central tradeoff: compressed remains stronger globally, while hybrid_state preserves slightly more state.

6.3 Paired Effects

Table 5 reports paired hybrid_state minus compressed deltas over matched experimental cells. Hybrid_state is lower on global quality ($\Delta Q = -0.208$, 95% CI $[-0.239, -0.177]$), but higher on state preservation ($\Delta = +0.066$, 95% CI $[+0.032, +0.100]$). This makes the EXP05 claim sharper: hybrid_state helps the operational-state variable, while compressed remains the safer global baseline.

Metric	n pairs	Δ	95% CI Low	95% CI High
quality index	774	-0.208	-0.239	-0.177
semantic fidelity	774	-0.174	-0.210	-0.138
state preservation	774	+0.066	+0.032	+0.100
operational continuity	774	-0.078	-0.112	-0.043
handoff quality	774	-0.199	-0.235	-0.162
compactness	774	-0.072	-0.118	-0.026
ambiguity	774	+0.463	+0.417	+0.510
information loss	774	+0.159	+0.121	+0.197

Table 5: EXP06 paired deltas for hybrid_state minus compressed. Positive deltas are beneficial for positive metrics but harmful for ambiguity and information loss.

6.4 Tokenizer/Variant Ablation

In the tokenizer ablation, native ZH remains stronger than romanized ZH_PINYIN (Table 6). This supports the hypothesis that script representation and tokenization interact with compact protocol behavior. It does not fully isolate tokenization from training distribution or language familiarity, but it is stronger evidence than the correlational ZH signal in EXP05.

Variant	n	Q	Fidelity	State	Stability	Compact
ZH	456	3.824	3.785	3.836	4.417	4.703
ZH_PINYIN	180	3.328	3.246	3.363	3.687	4.569
EN_LITERAL	180	3.595	3.506	3.650	4.341	4.617
ES_LITERAL	180	3.671	3.624	3.661	4.157	4.709

Table 6: EXP06 language/tokenizer ablation. Native ZH outperforms romanized ZH_PINYIN, suggesting that representation matters for compact protocol behavior.

6.5 Judge Calibration

Judge agreement is high for an LLM-as-judge setup (Table 7). Pairwise quality-index correlations range from 0.886 to 0.913. Azure GPT 5.4 High is stricter on average, Gemini 3.5 Flash is more generous, and Grok 4.20 Reasoning is intermediate. Because the same high-level conclusion survives three judges, EXP06 is less vulnerable to single-judge drift than EXP05 alone.

6.6 Interpretation

EXP06 strengthens the paper by converting EXP05’s broad exploratory signal into a controlled follow-up. The result is not that hybrid_state dominates compressed. Instead, compressed remains the general-purpose baseline, while hybrid_state selectively improves the operational-state variable

Judge A	Judge B	n	Pearson r	Mean Abs. Δ
Azure GPT 5.4 High	Gemini 3.5 Flash	1561	0.901	0.570
Azure GPT 5.4 High	Grok 4.20 Reasoning	1561	0.913	0.395
Gemini 3.5 Flash	Grok 4.20 Reasoning	1563	0.886	0.512

Table 7: EXP06 judge agreement over quality index.

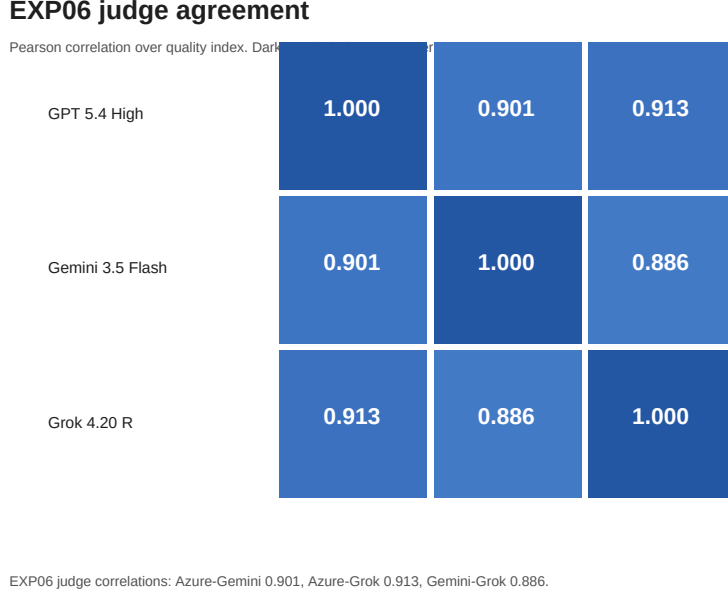


Figure 3: EXP06 judge-agreement matrix over quality index. Correlations are high enough to reduce single-judge risk, but not high enough to treat any judge as ground truth.

that motivated the protocol. This distinction makes the contribution sharper: symbolic protocols should be optimized and evaluated against their intended transfer function, not only against aggregate quality.

7 Real-Agent Handoff Evaluation: EXP07

EXP07 tests whether the protocol family can be evaluated in real agent frameworks using objective continuation metrics. This experiment is smaller than EXP05/EXP06, but it changes the evidence type: outputs are scored for explicit operational recovery rather than only by LLM judges. The cleaned dataset contains 270 successful executions: 180 LangGraph executions and 90 OpenFang executions.

7.1 Objective Metrics

Each EXP07 task defines a handoff state containing variables, constraints, completed subtasks, pending subtasks, plan steps, risks, validation conditions, and the next action. The continuation agent must return strict JSON. We then compute:

- variable recovery rate;
- subtask completion rate;
- plan continuity rate;
- constraint retention rate;
- binary handoff success;
- state-error count.

These metrics are intentionally narrower than EXP05/EXP06’s judge scores. They do not measure all semantic quality, but they provide a more objective view of whether operational state survived handoff.

7.2 Mode Results

Table 8 reports global EXP07 results by mode. Compressed remains the strongest operational baseline on variable recovery, plan continuity, and state-error count. Hybrid_state is slightly higher on constraint retention, but it does not win globally. Natural prose remains competitive on subtask completion and handoff success, but it produces more state errors.

Mode	n	VarRec	Subtasks	PlanCont	Constraints	Handoff	StateErr
natural	90	0.739	0.473	0.497	0.683	0.367	16.978
compressed	90	0.856	0.467	0.589	0.739	0.356	10.589
hybrid_state	90	0.750	0.364	0.442	0.744	0.344	10.811

Table 8: EXP07 real-agent objective metrics by mode. Higher is better except for state-error count.

7.3 Framework and Model Effects

Table 9 shows that framework surface matters. OpenFang reaches perfect scores under the current schema, while LangGraph exposes substantial variation across modes and models. We interpret the OpenFang result conservatively: it demonstrates compatibility between the protocol/task schema and an external real-agent surface, but should not be read as a universal ranking of OpenFang over LangGraph.

Framework	n	VarRec	Subtasks	PlanCont	Constraints	Handoff	StateErr
LangGraph	180	0.672	0.152	0.264	0.583	0.033	19.189
OpenFang	90	1.000	1.000	1.000	1.000	1.000	0.000

Table 9: EXP07 objective metrics by real-agent framework. OpenFang saturation should be interpreted as surface/schema compatibility, not as a general framework benchmark.

7.4 Paired Real-Agent Deltas

Table 10 reports paired `hybrid_state` minus compressed deltas. Within LangGraph, `hybrid_state` does not improve global operational recovery. With GPT chat latest, it improves constraint retention ($\Delta = +0.183$) and slightly reduces state errors ($\Delta = -0.133$), but loses subtask completion and plan continuity. With GPT 5.4 High, it is lower on most objective metrics. OpenFang is tied because all modes saturate the schema.

Framework	Model	n	Δ VarRec	Δ Subtasks	Δ Plan	Δ Constraints	Δ StateErr
LangGraph	GPT 5.4 High	30	-0.293	-0.125	-0.342	-0.167	+0.800
LangGraph	GPT chat latest	30	-0.025	-0.183	-0.100	+0.183	-0.133
OpenFang	GPT chat latest	30	0.000	0.000	0.000	0.000	0.000

Table 10: EXP07 paired deltas for `hybrid_state` minus compressed. Positive values are beneficial except for state-error count, where lower is better.

7.5 Interpretation

EXP07 supports the broader research program but narrows the claim. It shows that compact handoff protocols can be exercised in real agent frameworks and scored with objective state-continuation metrics. However, it does not replicate the stronger judge-based `hybrid_state` advantage as a global real-agent effect. The most conservative interpretation is that compressed remains the robust baseline, while `hybrid_state`’s state benefit is conditional: it may help explicit constraint retention in some real-agent/model strata, but it is not a universal operational winner.

8 Real-Agent Scale-Up: EXP08

EXP08 tests whether the EXP07 real-agent result survives a larger matrix. It uses 30 operational tasks, 2 protocol modes, 5 repetitions, and 3 framework/model routes: LangGraph with Azure GPT chat latest, LangGraph with Azure GPT 5.4 High, and OpenFang with Azure GPT chat latest. The operational-complete view contains 900/900 OK cells with no latest operational errors. Because strict JSON is part of the objective metric pipeline, we also report a stricter view containing 824/900 JSON-valid cells.

View / Mode	n	VarRec	Subtasks	PlanCont	Constraints	Handoff	StateErr
Operational / compressed	450	0.894	0.105	0.526	0.813	0.033	5.478
Operational / <code>hybrid_state</code>	450	0.756	0.053	0.493	0.750	0.020	14.344
JSON-valid / compressed	432	0.932	0.109	0.548	0.847	0.035	1.581
JSON-valid / <code>hybrid_state</code>	392	0.867	0.061	0.566	0.861	0.023	1.819

Table 11: EXP08 real-agent scale-up by analysis view and mode. Higher is better except for state-error count.

In the operational-complete view, compressed is the stronger global baseline across variable recovery, subtask completion, plan continuity, handoff success, and state-error count. In the stricter JSON-valid view, `hybrid_state` is slightly higher on constraint retention and plan continuity, but it still does not win globally. The most important new observation is a failure mode rather than a protocol win: Azure GPT 5.4 High under LangGraph produced 76 blank-output cells with

`parse_error=no_json_object`, including 58 in `hybrid_state` and 18 in `compressed`. These calls are operationally completed but not usable for strict schema scoring, showing that high-reasoning model routes can fail under JSON-constrained handoff even when the surrounding agent runner succeeds.

9 Discussion

The main result is not that a symbolic protocol simply beats natural language everywhere. Instead, EXP05, EXP06, EXP07, and EXP08 show tradeoffs between global quality, compactness, operational state, and real-agent continuation. `Compressed` remains the strongest universal baseline: it performs well across languages, models, controlled follow-up tests, and real-agent objective metrics. `Hybrid_state` gives up some global quality but improves state preservation in judge-based paired experiments, which is precisely the property that motivated the protocol. EXP07 and EXP08 add a caution: the `hybrid_state` advantage should not be generalized to every real-agent metric or framework.

This suggests a practical design rule: compact agent protocols should be optimized by target function. If the goal is general semantic quality, `compressed` is the safer baseline. If the goal is durable handoff and preservation of operational state, `hybrid_state` is more promising. If the goal is extreme brevity, `hybrid_min` is compact but risky.

The ZH result is one of the most interesting signals. Chinese performs best in the EXP05 principal language comparison and remains stronger than ZH_PINYIN in EXP06-B. A plausible explanation is that script density, tokenizer behavior, and model training distribution interact with symbolic compression. However, even EXP06 cannot fully separate these factors. The safest claim is that representation matters; the exact mechanism remains open.

Judge drift also matters. The same outputs can receive systematically different quality levels depending on evaluator family. This does not invalidate the result, but it means future papers should report judge identity, judge agreement, and sensitivity analysis.

9.1 Claim Strength

The strongest claim is that `compressed` is the most reliable general-purpose baseline in this experimental family. It wins quality in EXP05, remains ahead in EXP06 under stricter paired controls, and is also the best aggregate operational baseline in EXP07 and EXP08 real-agent objective metrics.

The second strong claim is narrower: `hybrid_state` improves state preservation under judge-scored paired designs. This appears in EXP05’s principal paired analysis and survives EXP06’s stricter paired design. The claim should not be overstated as a global quality victory or a universal real-agent operational win.

The third claim is methodological: real-agent handoff can be measured with objective state-continuation metrics. EXP07 demonstrates this feasibility in LangGraph and OpenFang, and EXP08 scales it to 900 operational executions. The substantive result is mixed: `compressed` remains stronger globally, while `hybrid_state` shows narrower constraint/plan benefits only in selected strata.

The ZH/tokenization claim is promising but still partly exploratory. EXP05 shows a ZH advantage, and EXP06-B shows native ZH outperforming romanized ZH_PINYIN. This supports a representation effect, but not a complete causal explanation.

The weakest claims are any broad model-intelligence rankings. Generator effects in this paper reflect protocol/task/provider behavior, not general model capability.

10 Limitations

EXP05 is exploratory and systems-oriented, while EXP06 is a narrower controlled follow-up but still depends on LLM judges and provider serving layers. EXP07 and EXP08 reduce judge dependence through objective metrics, but their framework surfaces are not interchangeable. The use of LLM judges introduces evaluator bias, even though EXP06 reduces single-judge risk through three-judge calibration. Some models were accessed through provider-specific serving layers, so observed behavior includes provider routing, quota, truncation, latency, safety filtering, and content-policy behavior. In EXP06, 21 Azure GPT instant generations were retained as terminal policy failures and excluded from score aggregation; therefore Azure generator averages should be read with survivorship bias in mind. In EXP08, 76 Azure GPT 5.4 High cells returned blank outputs under strict JSON scoring; these are retained as a model/serving failure mode rather than silently removed.

The quality index is descriptive, not a validated psychometric scale. It is useful for ranking and paired contrasts, but the paper’s main scientific claims rely on individual metrics such as state preservation, operational continuity, compactness, ambiguity, and information loss. The ZH effect is promising but not fully causal. EXP06-B strengthens the representation hypothesis by comparing ZH with ZH_PINYIN and literal variants, but it still cannot isolate tokenizer mechanics from training distribution and language familiarity. Some mini-experiments use smaller samples and should be treated as hypothesis-generating.

The model list is also time-sensitive. Frontier model names, availability, quotas, and pricing change rapidly. Therefore, this paper should describe exact observed model identifiers, dates, and serving routes in an appendix before publication. EXP07’s OpenFang results saturate the objective schema, and EXP08’s OpenFang results remain high, so they should be interpreted as successful compatibility checks rather than general framework benchmarks.

11 Conclusion

EXP05, EXP06, EXP07, and EXP08 together reframe prompt compression as interoperable state-transfer protocol design. EXP05 provides the broad discovery result: compressed is a strong cross-model, multilingual baseline, and hybrid_state improves state preservation in paired analysis. EXP06 provides the controlled follow-up: compressed remains ahead on global quality, but hybrid_state still improves state preservation under stricter paired controls. EXP07 adds real-agent validation, and EXP08 scales it: compact protocols can be executed and measured in LangGraph and OpenFang using objective handoff metrics, but hybrid_state does not dominate globally in this setting. The conclusion is therefore not that one protocol dominates all uses. Rather, compressed is the universal baseline, and hybrid_state is a specialized state-oriented protocol whose benefits are conditional and should be tested against the target execution surface.

This distinction makes the research program clearer. Future compact agent protocols should be evaluated by target function: general semantic quality, state preservation, continuity, recoverability, cost, framework behavior, and robustness to language and model family. The results motivate further work on symbolic protocols for multi-agent AI systems, especially where agents must survive context loss, provider substitution, multilingual transfer, evaluator drift, and real-agent execution constraints.

References

- [1] Anonymous Authors. EXP06 Causal Protocol Transfer: Frozen Data, Cleaned Dataset, and Statistical Reports. Internal experimental dataset freeze, May 26, 2026.
- [2] Anonymous Authors. EXP05 Premium Multi-Model Trilingual: Cleaned Experimental Data and Reports. Internal experimental dataset freeze, May 23, 2026.
- [3] Anonymous Authors. EXP07 Real-Agent Handoff Objective Metrics: Frozen Data, Cleaned Dataset, and Statistical Reports. Internal experimental dataset freeze, May 26, 2026.
- [4] Anonymous Authors. EXP08 Real-Agent Scale-Up: Frozen Data, Objective Metrics, and Blank-Output Failure Analysis. Internal experimental dataset freeze, May 27, 2026.
- [5] Jason Wei et al. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. arXiv:2201.11903, 2022.
- [6] Shunyu Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629, 2023.
- [7] Nelson F. Liu et al. Lost in the Middle: How Language Models Use Long Contexts. arXiv:2307.03172, 2023.
- [8] Yupeng Chang et al. A Survey on Evaluation of Large Language Models. ACM Transactions on Intelligent Systems and Technology, 2024.